# Web3 To-Do Contract

## Project Summary

This project is a decentralized task management application built on the Ethereum Sepolia testnet. Unlike traditional to-do apps that store data on centralized servers, this DApp stores all tasks directly on the blockchain through a Solidity smart contract called `TodoContract.sol`.

### How it works:

Any user with an Ethereum wallet can connect and interact with the contract. Each task is permanently linked to the wallet address that created it, meaning no one else can modify or delete it. There is no login, no password, and no central authority, the blockchain handles identity automatically through wallet signatures.

The contract supports full CRUD functionality:

- Create :  add a new task, stored on-chain with the caller's wallet as owner
- Read : fetch a single task or all tasks belonging to your wallet
- Update : toggle a task between pending and completed, or edit its content
- Delete : soft-delete by resetting the struct to default values, since blockchain data is immutable

### Key design decisions:

Ownership is enforced through a `require` check on every write function, so only the task creator can modify their own tasks. Gas efficiency is maintained by using `uint256` (the EVM's native word size) for IDs and capping task content at 280 bytes. The contract emits events (`TaskCreated`, `TaskToggled`, `TaskEdited`, `TaskDeleted`) so all activity is publicly verifiable on Sepolia Etherscan.

The contract is deployed at `0xdd558475249E4caac26409D6970b9fD09BC9Df21` on Sepolia, with the frontend demo built in HTML/JS simulating the full on-chain interaction flow.